

Agentic AI for Knowledge Production

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Today's Agenda

Gentzkow and Shapiro (2014) revisited

Tests

Gentzkow and Shapiro (2014)

revisited

Automate everything that can be automated

Write a single script that executes all code from beginning to end

Store code and data under version control

Run the whole directory before checking it back in

Separate directories by function

Separate files into inputs and outputs

Make directories portable

Abstract to eliminate redundancy

Abstract improve clarity

Otherwise, don't abstract

Don't write documentation you will not maintain
Code should be self-documenting

Manage tasks with a task management system
Email is not a task management system

Tests

Tests are **VERY** important.

Reinhart and Rogoff (2010)

Five countries dropped from one average.

−0.1% published. +2.2% corrected.

One row count catches it.

The agent games the check

Hardcodes the answer: 33% of the time.

Fakes a solution: 22%.

(EvilGenie, ambiguous specs. Yours are ambiguous.)

If you're not reading code, write the check.

Four checks

Check	When	Example
Data assertion	Every merge	Rows in = rows out. <code>isid id year</code> .
Known answer	Closed form	SVG to lat/lon: round-trip the four corners.
Parameter recovery	No answer	BLP: simulate from β , get β back.
Coverage	Randomness	Bootstrap: 95% interval covers truth 95% of the time.

None apply? Test what cannot change. Shuffle rows, coefficients hold. Weights of one, unweighted run.

What you type

```
# tests/test_clean.py      $ pytest
def test_merge_keeps_every_parcel():
    out = merge_owners(parcel, owners)
    assert len(out) == len(parcel)

def test_boot_se_matches_ols():
    rng = np.random.default_rng(42)
    np.testing.assert_allclose(
        boot_se(X, y, rng), ols_se(X, y),
        rtol=0.05)
```

test_*.py, test_*, assert, run
pytest. That is the framework.

Stata: assert, isid.

R: testthat.

Seed anything random.

Never == on floats.

1,000 rows, not the full run.

Freeze Table 3 and diff it.

What you tell the agent

Green checkmark

```
Write the BLP estimator  
and add some tests.
```

Checked estimator

```
Write the test first, then stop.  
Simulate 500 markets from  
beta = [-2, 1, 0.5]. Assert we  
recover it within 0.05.  
Show me the test.
```

You approve the test. The agent writes the code.

Say it in the prompt: never edit a test to make it pass.

Break the code once. A test that never fails is testing nothing.

Gentzkow and Shapiro, *Code and Data for the Social Sciences*. `pyblp/tests/test_blp.py`.